Nicholas James Fettig

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EDUCATION

University of Michigan | Ann Arbor, MI

M.S. Robotics — ***Aug 2026 - Dec 2027***

Current Coursework: *Math for Robotics, Robotic Systems Lab, Digital Twin Synchronization (research)*

Swarthmore College | Swarthmore, PA

B.S. Engineering, B.A. Computer Science | GPA 3.86 (3.96 in-major) — ***May 2026***

Honors & Awards: *Joshua Lippincott Graduate Fellow, Tau Beta Pi, Martin S. Kapp Scholarship*

TECHNICAL SKILLS

Languages: C++, C, Python, MATLAB, Bash, JavaScript/TypeScript, PostgreSQL

Robotics: ROS 2, EtherCAT (Beckhoff/TwinCAT), real-time Linux, hardware-in-the-loop (HITL) testing

Tools: CMake, Git, GDB, Valgrind, Docker, PyTorch/TensorFlow, AWS, Terraform, LaTeX

WORK EXPERIENCE

Shinkei Systems | Los Angeles, CA — ***May - August 2026***

Robotics Software Engineering Intern

- Integrated load-cell force sensors, read through two Beckhoff terminals, into an in-house motor-control service over EtherCAT; wrote SDO configuration for startup calibration and per-channel scaling
- Built a per-fish weight-extraction algorithm (ETL job) over the continuous load-cell weight stream, segmenting individual fish from the running signal to reach **<1% mean signed error at batch scale**

Reliable Robotics | Mountain View, CA — ***May - August 2025***

Software Engineering Intern

- Led development of "Big Board," a Mission Control display used by remote pilots operating automated aircraft, integrating live ADS-B telemetry for **8 company aircraft** in a safety-critical operational loop (LitJS, AWS EC2)

Carnegie Mellon University | Pittsburgh, PA — ***May 2024 - May 2025***

Computer Science Undergraduate Researcher

- Surveyed and classified **200+ works** on RL-based traffic signal control into a novel taxonomy of deployment challenges (sim-to-real transfer, safety constraints, infrastructure limitations)

PROJECTS

Autonomous Mobile Robot | *Raspberry Pi, ROS 2, C++, ESP32*

- Built a mecanum-drive mobile robot from scratch: RPLIDAR A1M8 and MPU6050 IMU into an ROS 2 stack on a Raspberry Pi 5, with an ESP32-S3 handling closed-loop motor control and wheel odometry over serial

SEPTA Transit Display | *Raspberry Pi, Python, HUB75*

- Built a 256x128 LED matrix display (16 chained HUB75 panels) for live SEPTA train arrivals, solving panel chain addressing and coordinate mapping for the 4x4 grid; deployed as a systemd service on a Pi 5

Fish Species Classification | *TensorFlow, ResNet50, Computer Vision*

- Trained a ResNet50 classifier with early fusion, concatenating segmentation masks as a fourth input channel alongside RGB, achieving **97.22% classification accuracy**

Home Server | *Docker, Linux, Cloudflare*

- Run an Ubuntu home server hosting containerized services (personal site, blog, Jellyfin) behind key-only SSH and a Cloudflare tunnel